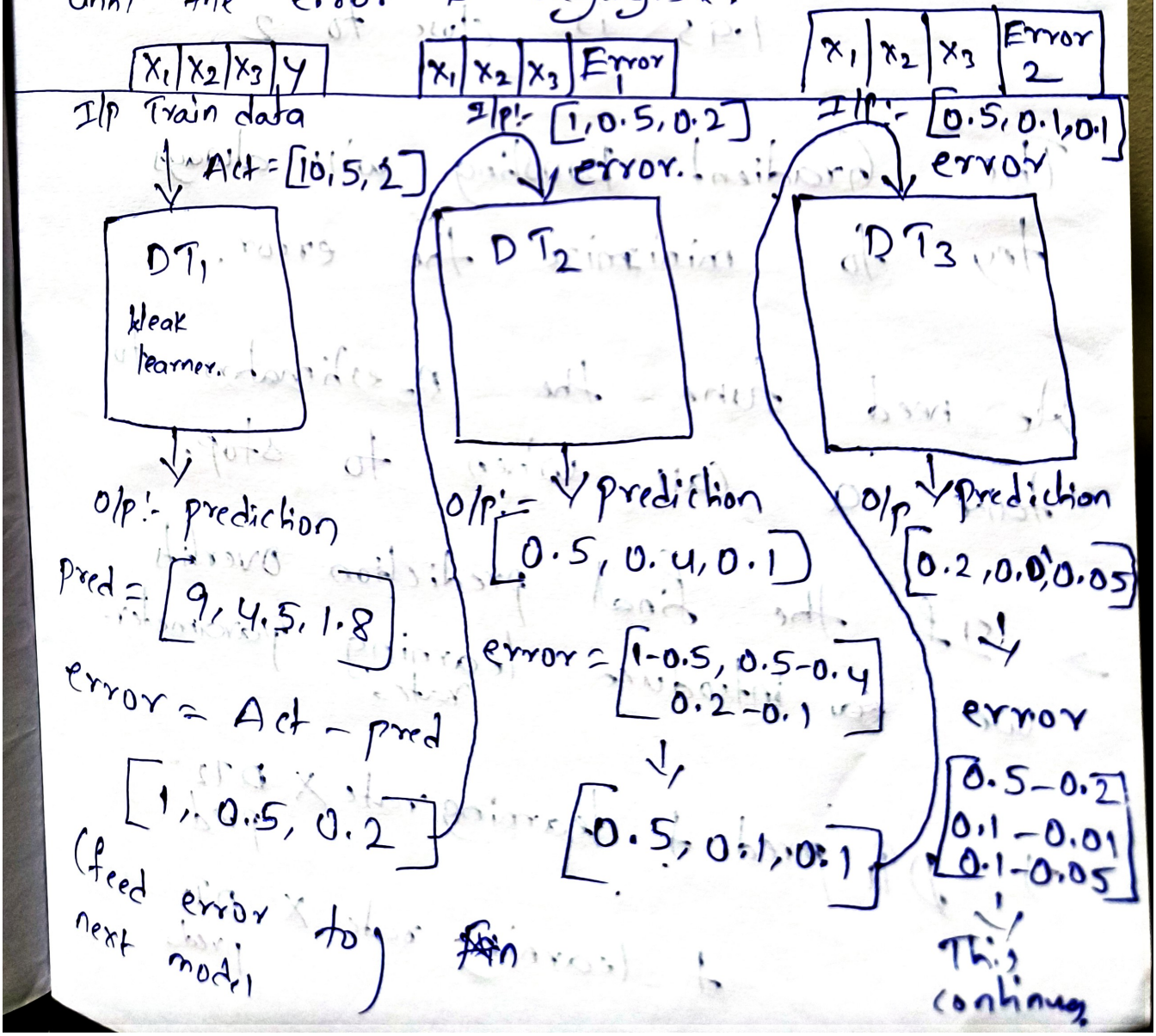
9. Gradient Boosting.

→ Basically in Gradient Boosting, first we will consider a learner (weak). Now the weak learners will do the predictions and the we are going to calculate the error (Actual - pred) and the we will pass these errors to the next learner. this iteration continues until the error is negligible.



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Final o/p = sum of all predictions.

DT1 DT2 DT3

$= [9, 4.5, 8] \; [0.5, 0.4, 0.1] \; [0.2, 0.01, 0.05]$

$= \begin{bmatrix} 9 + 0.5 + 0.2 \\ 4.5 + 0.4 + 0.01 \\ 1.8 + 0.1 + 0.05 \end{bmatrix} \Rightarrow \begin{bmatrix} 9.7 \\ 4.91 \\ 1.95 \end{bmatrix}$

Now, 9.7 is close to 10
4.91 is close to 5
1.95 is close to 2.

The Gradient Boosting will always try to minimize the error.

We need tune the n_estimators to take a call when to stop.

→ If the final prediction overfits we introduce learning parameter. (learning rate)

→ DT1 pred. + learning rate X DT2 pred.
+ learning rate X DT3 pred.

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⑧. XGBoost

Xtreme Gradient Boosting => Gradient Boosting + Regularisation

⑨. Voting classifier :- (soft voting / Hard voting).

Group of different models.

Hard voting.

Knn	RF	GB
1	0	1

prediction = mode [1, 0, 1]

= 1

In hard voting mode of all Models prediction is taken

soft voting

Knn	RF	GB
1 → 0.55	0.1	0.51
0 → 0.45	0.9	0.49

so where probability will be is considered.

overall class - 1 probability

(0.55 + 0.1 + 0.51)

= 1.07

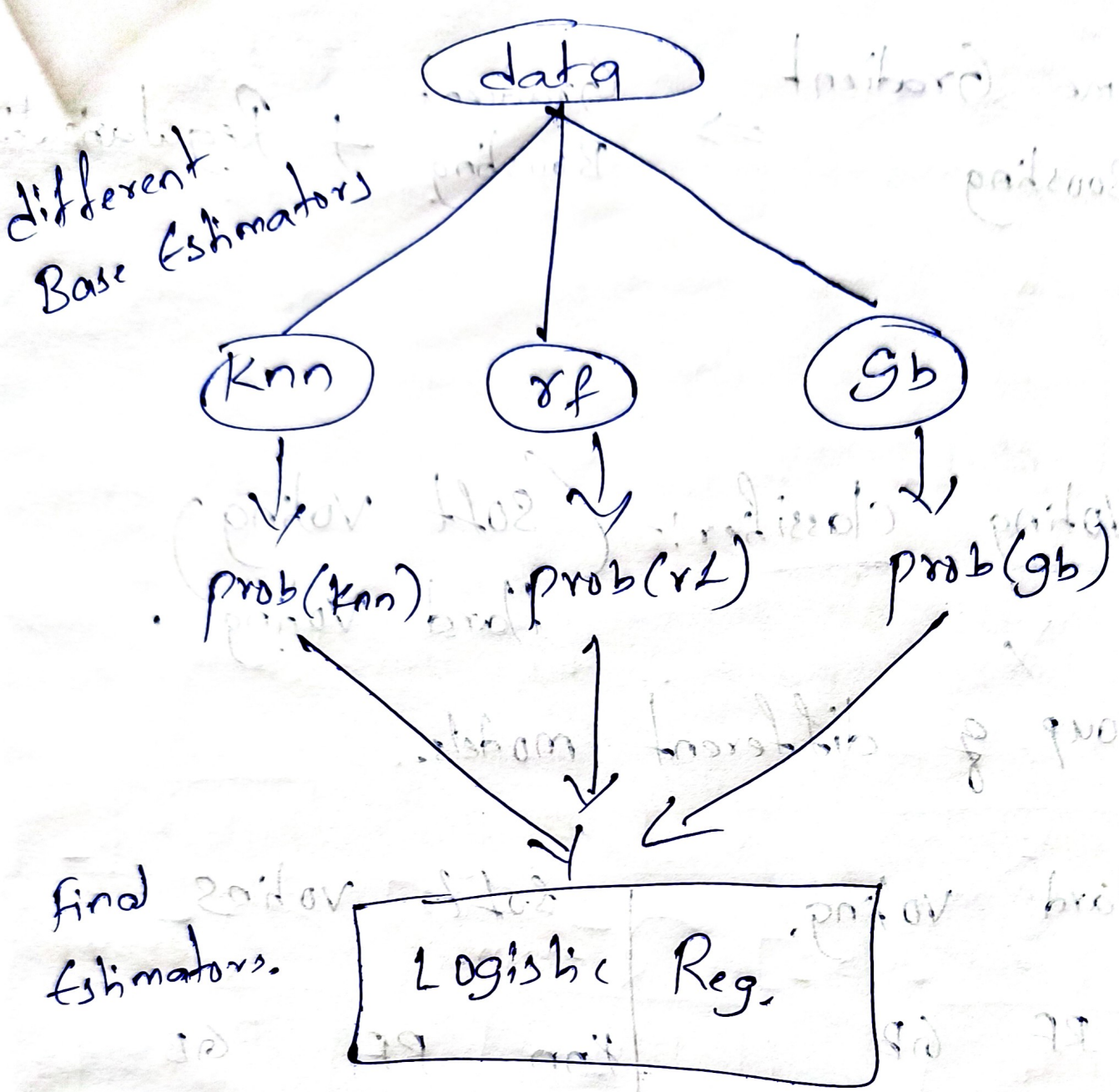
overall class - 2 prob

(0.45 + 0.9 + 0.49) = 1.84

now the prediction will be 0

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Stacking Classifier



The probabilities of base estimators are passed to final estimators where final prediction is made.

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